

JOSE TUPAYACHI

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EDUCATION

University of Tennessee, Knoxville

Ph.D., Industrial Engineering May 2024 - May 2027

M.S., Industrial Engineering Aug 2022 - May 2024

Pontifical Catholic University of Peru

B.S., Industrial Engineering March 2014 - Feb 2020

EXPERIENCE

Developer

Mar 2025 - Present

Oak Ridge National Laboratory

Oak Ridge, TN

- Developed an AI/ML pipeline fusing spectral and spatial Atom Probe Tomography (APT) data with a feedback-driven loop for nuclear materials characterization to remove user bias and automate analysis.
- Fine-tuned vision transformers within a multi-pathway hybrid architecture for automated screening of mesoscale weather station data, classifying wind speed and direction for rapid real-time emergency atmospheric response projections of plume impacts.
- Deployed explainable Machine Learning models utilizing large-scale atmospheric and geospatial datasets for predictive analytics.
- Developed an automated geospatial dashboard platform integrating natural language processing, knowledge-graph reasoning, and interactive spatial analytics.
- Automated experiment parameter selection by processing 1,000+ SEM images and X-ray diffraction (XRD) spectra for microstructure analysis.
- Engineered a Retrieval-Augmented Generation (RAG) system using LangChain and LangGraph with Qwen/Llama models to ingest and cross-reference 200+ lithium breeder blanket research papers, accelerating literature review cycles while maintaining traceability.

Graduate Research Assistant

Jul 2024 - Present

University of Tennessee, Knoxville

Knoxville, TN

- Led the development of an AI-powered freight transportation digital twin, linear optimization, realtime routing and parallel processing for the DOE ARPA-E.
- Designed and coded end-to-end mobile health applications (SmartShots, ACTAPP), publishing Android, iOS, watchOS, and Wear OS mobile distributions with secure integration of patient data.
- Designed an intermodal routing optimizer integrating diverse data sources (road, rail, waterway) using ontology-guided GIS, with REST API integration and complex workflow orchestration.
- Developed TW-GCN (Traffic-Weather Graph Convolutional Network) framework for EV charging demand forecasting, feeding high resolution traffic message channels (TMC) into stacked architectures to achieve superior predictive accuracy over classical ML models.

Data Engineer

Mar 2022 - Aug 2022

Indra

- Built and maintained scalable ETL/ELT pipelines using Python, Scala, and Shell scripting on UNIX/Linux, processing high-volume datasets in Agile delivery cycles.
- Developed distributed data processing systems using Scala/PySpark on Apache Spark, Hadoop, Hive/HQL, and AWS (S3, EMR), migrating a legacy Oracle warehouse to a partitioned cloud lakehouse and tuning Spark memory and resource allocation for efficient job execution.
- Designed dimensional data models for data warehousing and implemented entity-resolution and deduplication logic to unify records across disparate tables.
- Implemented CI/CD deployment automation with Jenkins for automated builds, unit testing, and application deployment including live deployment, rollback, and backups.

Business Intelligence Analyst

Jan 2022 - Mar 2022

Globokas

- Developed executive dashboards and KPI reporting systems supporting strategic and operational decision-making for multiple business clients.
- Integrated heterogeneous data sources (REST APIs, internal systems, external platforms) into unified analytics pipelines

Data Analyst

Nov 2020 - Dec 2021

Enel Distribucion

- Built interactive Power BI and Tableau dashboards for operational monitoring and KPI reporting.
- Managed SQL/T-SQL databases and Salesforce integrations via API queries.
- Developed a PyQt desktop application to automate invoice verification and special-case processing.
- Engineered customer segmentation models using unsupervised machine learning (K-Means and hierarchical clustering) on payment history and demographics, enabling collection teams to prioritize outreach and improve payment-recovery efficiency.

Developer - Portfolio Valuation

Aug 2020 - Nov 2020

GPS Management

- Implemented machine learning models for non-performing loan (NPL) recovery and portfolio valuation.
- Automated scoring workflows, reducing manual processing time.

Professional IT Intern

Sep 2020 - Oct 2020

Superintendencia de Banca, Seguros y AFP

- Assisted in database maintenance and internal reporting for financial supervision operations.
- Supported IT infrastructure and regulatory data management

IT Intern

Feb 2019 - Aug 2019

Administradora Clinica Ricardo Palma

- Developed and maintained healthcare plans information systems.
- Supported electronic health record integration and database administration for clinical operations.

CLIENTS & SPONSORS

Oak Ridge National Laboratory - Materials & Multimodal AI

Mar 2025 - Present

Developer

- Funding: DOE Office of Science (SC), Fusion Energy Sciences (FES), DOE Office of Environmental Management (EM), Nuclear Safety Research and Development (NSRD). <https://demo2.recoil.ise.utk.edu> ; <https://github.com/jtupayachi/smartgiscavp>

RECOIL - Cognitive Freight Transportation Digital Twin

Jul 2024 - Mar 2026

Lead AI/Data Engineer

- Funding: U.S. Department of Energy ARPA-E, Project #DE-AR0001780. <https://m1.recoil.ise.utk.edu>

SmartShots - Childhood Vaccination App

Dec 2023 - Dec 2024

Full-Stack Developer

- Winner, IISE DAIS Student Mobile App Competition 2024. Funding: Tennessee Department of Health. <https://www.iise.org/Details.aspx?id=33697> ; <https://play.google.com/store/apps/details?id=com.ilab.smartshots>

Active Caregiver's Toolkit (ACTAPP) - Mobile Health App

Jul 2024 - Present

Mobile Developer

- Funding: Hillman Emergent Innovation (HEI). https://play.google.com/store/apps/details?id=com.ilabutk.fe_rich ;

Variable Matrix - For Analytics

2022

Python Developer

- Client: For Analytics (Contracted by Indra) - Banco de Credito del Peru <https://www.viabcp.com>

SELECTED PUBLICATIONS

- Wyatt, T. H., Lowe, S., Tupayachi, J., & Li, X. (2026). Design and usability testing of SmartSHOTS: A mobile app to reduce vaccine barriers for children 0-24 months. *Health Informatics Journal*, 32(1). <https://doi.org/10.1177/14604582261419334>
- Tupayachi, J., Xu, H., Omitaomu, O. A., Camur, M. C., Sharmin, A., & Li, X. (2024). Towards next-generation urban decision support systems through AI-powered construction of scientific ontology using large language models. *Smart Cities*, 7(5), 2392-2421. <https://doi.org/10.3390/smartcities7050094>
- Tupayachi, J., Ferguson, M. M., & Li, X. (2024). A simulation-based real-time deep reinforcement learning approach for fighting wildfires. *Proceedings of the 2024 Annual Modeling and Simulation Conference (ANNSIM)*, 1-12. <https://doi.org/10.23919/annsim61499.2024.10732085>
- Xu, H., Li, X., Tupayachi, J., Lian, J. J., & Omitaomu, O. A. (2024). Automating bibliometric analysis with sentence transformers and RAG. *Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Advances in Urban-AI*, 1-10. <https://doi.org/10.1145/3681780.3697252>
- Li, X., Tupayachi, J., Sharmin, A., & Martinez Ferguson, M. (2023). Drone-aided delivery methods, challenges, and the future: A methodological review. *Drones*, 7(3), Article 191. <https://doi.org/10.3390/drones7030191>
- Khan, A. N., Tupayachi, J., Dabbs, W., Yao, B., & Chamberlin, S. (in press). CliniSense AI for automated clinical skills assessment with real-time feedback. *npj Digital Medicine (Accepted)*.
- Tupayachi, J., & Silva, L. (2022). Better efficiency on non-performing loans debt recovery and portfolio valuation using machine learning techniques. In *Production and Operations Management: POMS Lima, Peru* (pp. 33-53). Springer. https://doi.org/10.1007/978-3-031-06862-1_3
- Tupayachi, J., Yu, X.-Y., Li, X., Liu, Z., & Birdwell, K. (2026). *Explainable machine learning for extreme weather event prediction at the Oak Ridge Reservation* (SSRN Scholarly Paper No. 6409258). SSRN. <https://doi.org/10.2139/ssrn.6409258>
- Tupayachi, J., Khan, A. N., & Li, X. (2026). Scalable decentralized prognostics for industrial systems under data heterogeneity. *Computers & Electrical Engineering*, 133, Article 111023. <https://doi.org/10.1016/j.compeleceng.2026.111023>
- + Publications: <https://scholar.google.com/citations?user=Kyn6Et8AAAAJ&hl=en&oi=ao>

SKILLS

AI and Machine Learning

- Machine Learning Engineering, LLMs, RAG, Agentic AI Workflows, CNNs, Vision Transformers, TensorFlow, PyTorch, Statistical Modeling, Predictive Modeling, Clustering, Regression, Tree-Based and Boosting Models (Random Forest, LightGBM, XGBoost), Model Tuning and Calibration, Explainable AI

LLM and Agentic Frameworks

- LangChain, LangGraph, LiteLLM, vLLM, Ollama, QLoRA Finetuning

Cloud and Infrastructure

- AWS (S3, EMR, Redshift, SageMaker, ECS), Microsoft Azure, Google Cloud, Docker, Kubernetes, UNIX/Linux, KVM, Slurm, CI/CD, Jenkins, Git, GitLab

Data Engineering

- Python, SQL, Scala, Apache Spark, Hadoop/MapReduce, Hive, Kafka, ETL/ELT, Batch and Real-Time Streaming, Delta Lake, Databricks Workflows, Unity Catalog, dbt, Airflow, Prefect, Data Modeling, Entity Resolution, Data Quality, Anomaly Detection, Snowflake, Amazon Redshift, Dimensional Modeling, Lakehouse and Data Warehouse Architecture, Query and Performance Optimization

Databases

- PostgreSQL, MySQL, SQL Server, MongoDB, Cassandra

Programming Languages

- Java, R, SAS, STATA, Bash, Dart, PHP, C++, CUDA, SPSS

Web, Frameworks, and APIs

- Django, Flask, Streamlit, Dash, React, Flutter, Laravel, REST and GraphQL APIs

Optimization and Simulation

- Gurobi, CPLEX, AnyLogic, NetworkX

Geospatial and Project Management

- WMS, GeoServer, OverPass API, Knowledge-Graph Reasoning, Data-Privacy-Aware Design, JIRA, Confluence, Agile/Scrum

Languages

- English, Spanish, Portuguese

CERTIFICATIONS

- Databricks
- MS Transact-SQL (T-SQL)
- Microsoft Fabric DP-600 & DP-700
- Microsoft Fabric
- AWS
- GCP

AWARDS & HONORS

- IISE Future Faculty Fellows (2025-2026), USA - Awarded
- Best Paper Award, CIE51 (51st Conference on Computers and Industrial Engineering), Sydney, Australia (2024)
- IISE DAIS Student Mobile App Competition - Winner, SmartShots Project, Montreal (2024)
- HIDA Helmholtz Visiting Researcher, Karlsruhe Institute of Technology (KIT), Germany - Awarded (2024)
- Holiday Fellowship, University of Tennessee, Knoxville (2022, 2023, 2024)

TEACHING EXPERIENCE

University of Tennessee, Knoxville

Aug 2022 - Dec 2022
Knoxville, TN

- IE406/408: Simulation - Fall 2022
- IE483: Reliability Engineering - Spring 2023
- IE 533: Theory and Practice of Engineering Management - Fall 2026